

Individual Honours Project

WEEKLY REPORT

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UIT - BCU

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I. Summary

- This is the report of my project “**BCG: A Game-Based Approach to Motor Imagery Adaptation and BCI Calibration.**”
- The target was to make EEG-based motor imagery calibration more effective and easier to apply outside the laboratory.

I. Finished tasks

1. Model training ^[STP7] – 100%

STRATEGY 1

Classical ML on MIMED dataset

- Initial approach using CSP (Common Spatial Pattern) with classical classifiers. MIMED was chosen as the primary dataset because it was recorded on the EMOTIV Epoc X.
- However, the dataset suffers from very few trials per subject (~8), which causes unstable CSP covariance estimation and poor generalization.

Model	Classes	Protocol	Accuracy
CSP + LDA	4-class	Cross-subject, 5-fold	52.5%
CSP + SVM	4-class	Cross-subject, 5-fold	46.7%
CSP + LDA	4-class	Cross-subject, 5-fold	42.1%

I. Finished tasks

1. Model training ^[STP7] – 100%

STRATEGY 2

Deep Learning (EEGNet architecture) trained on BCI IV 2a dataset -> fine-tuned on MIMED

- To overcome the data scarcity problem, EEGNet was first pretrained on BCI IV 2a, a larger research-grade dataset with 5,184 trials across 9 subjects. The model learned general EEG patterns (mu/beta rhythms) from this data.
- The pretrained model was then adapted to Epoc X hardware by fine-tuning on MIMED.

Model	Classes	Protocol	Accuracy
EEGNet + BCI IV 2a	4-class	5-fold CV	61.44%
EEGNet + MIMED	2-class	LOTO	71.67%
EEGNet + BCI IV 2a then finetuned on MIMED	2-class	LOTO	72.08%

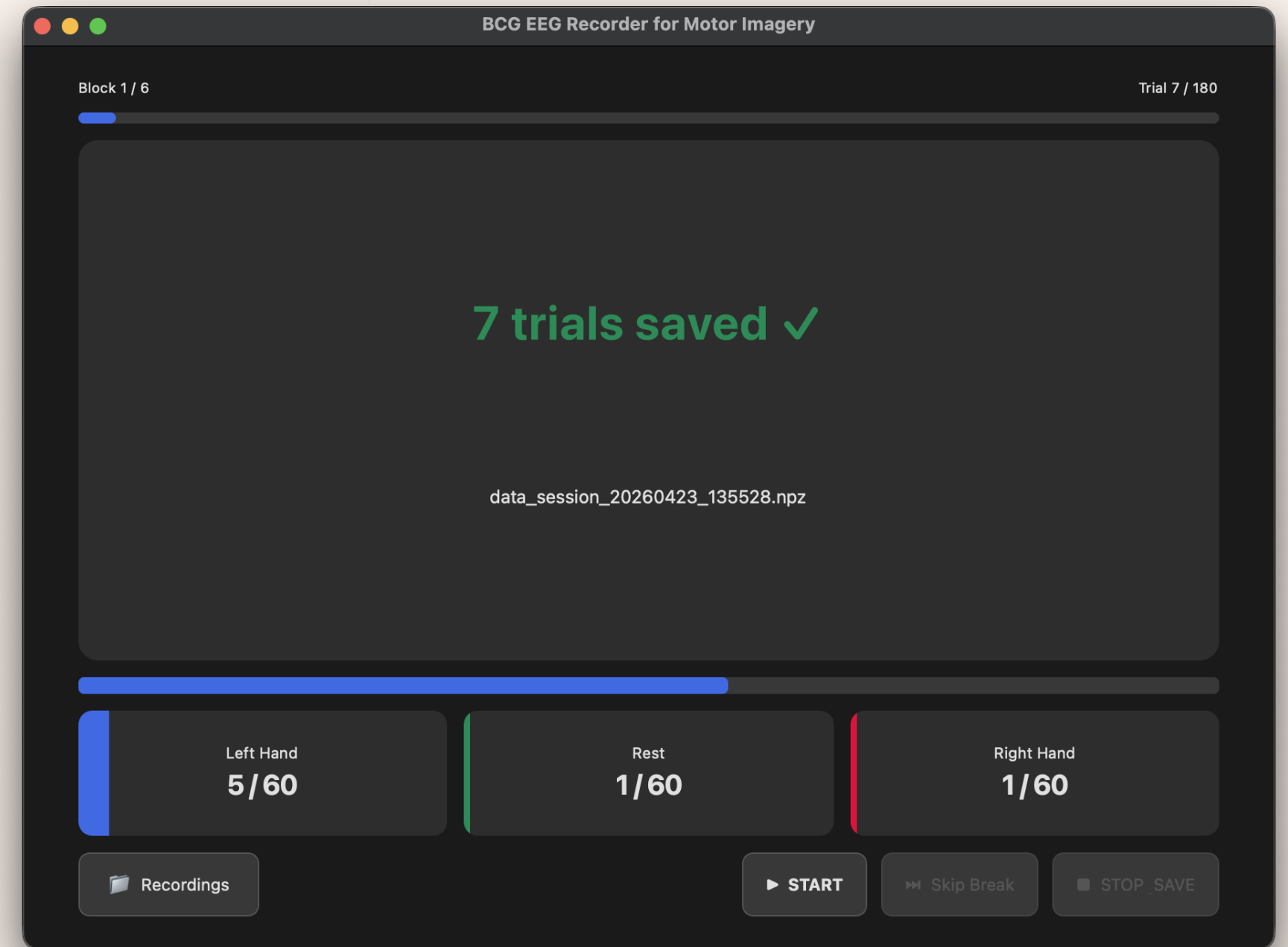
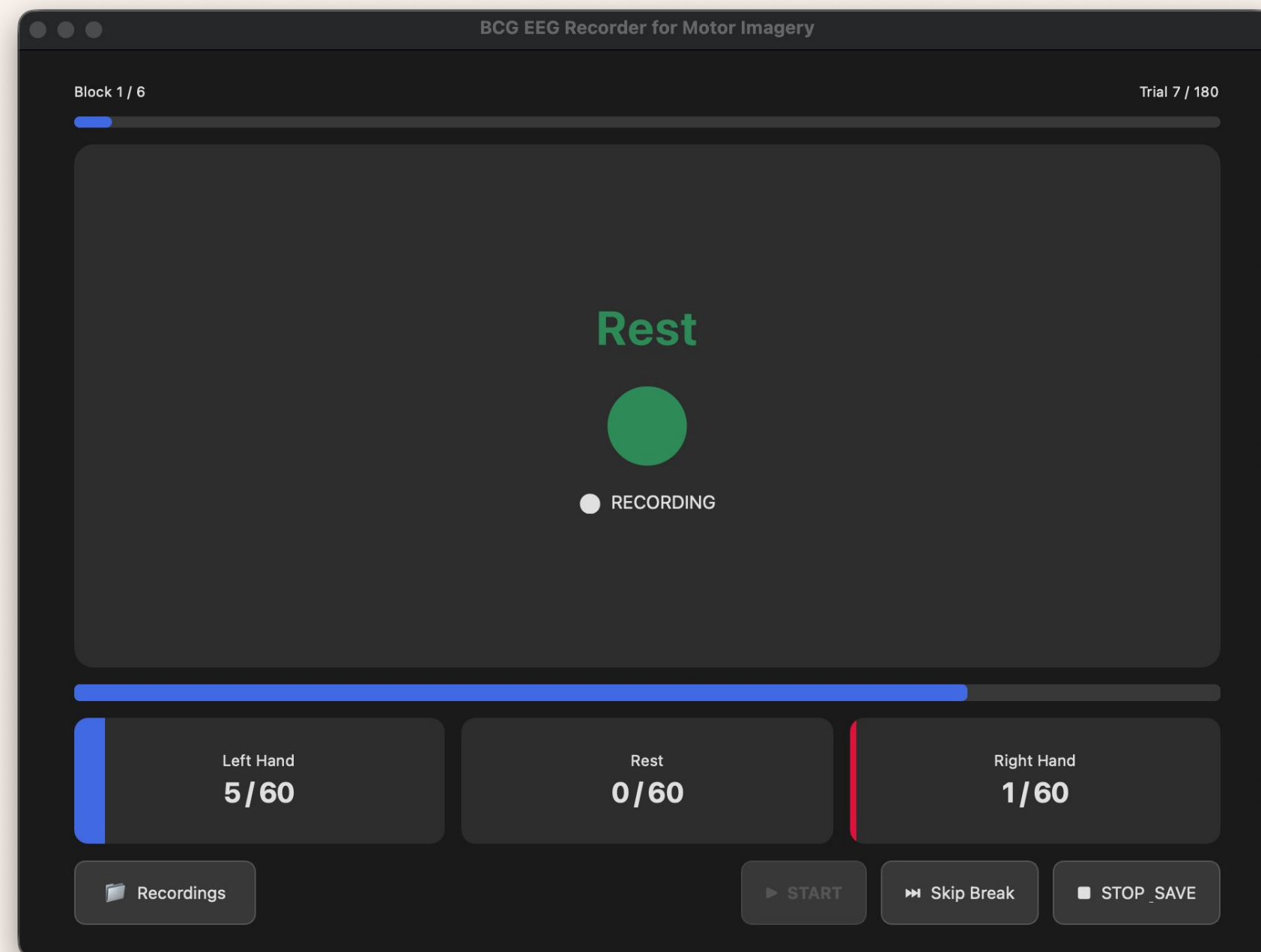
I. Finished tasks

2. Traditional Calibrating Method Design ^[STP11] - 100%

- I have implemented 2 desktop applications to support the experimental pipeline
- The **Data Collection Tool** guides participants through a cued EEG recording protocol across configurable blocks and trials, automatically saving labeled data for model training.
- The **Live Monitor Tool** performs real-time motor imagery classification by streaming EEG data into a sliding 4-second window and continuously displaying predictions.

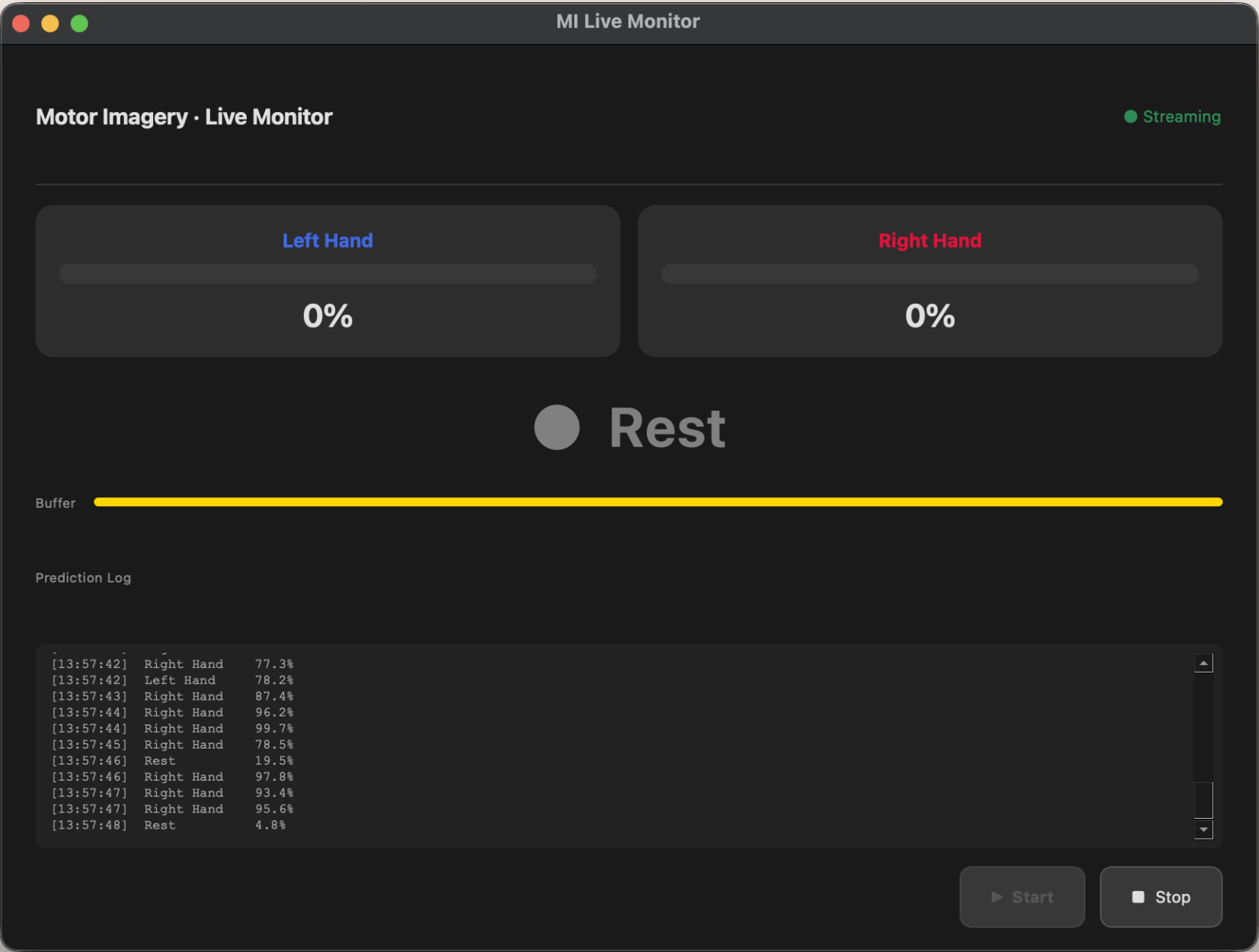
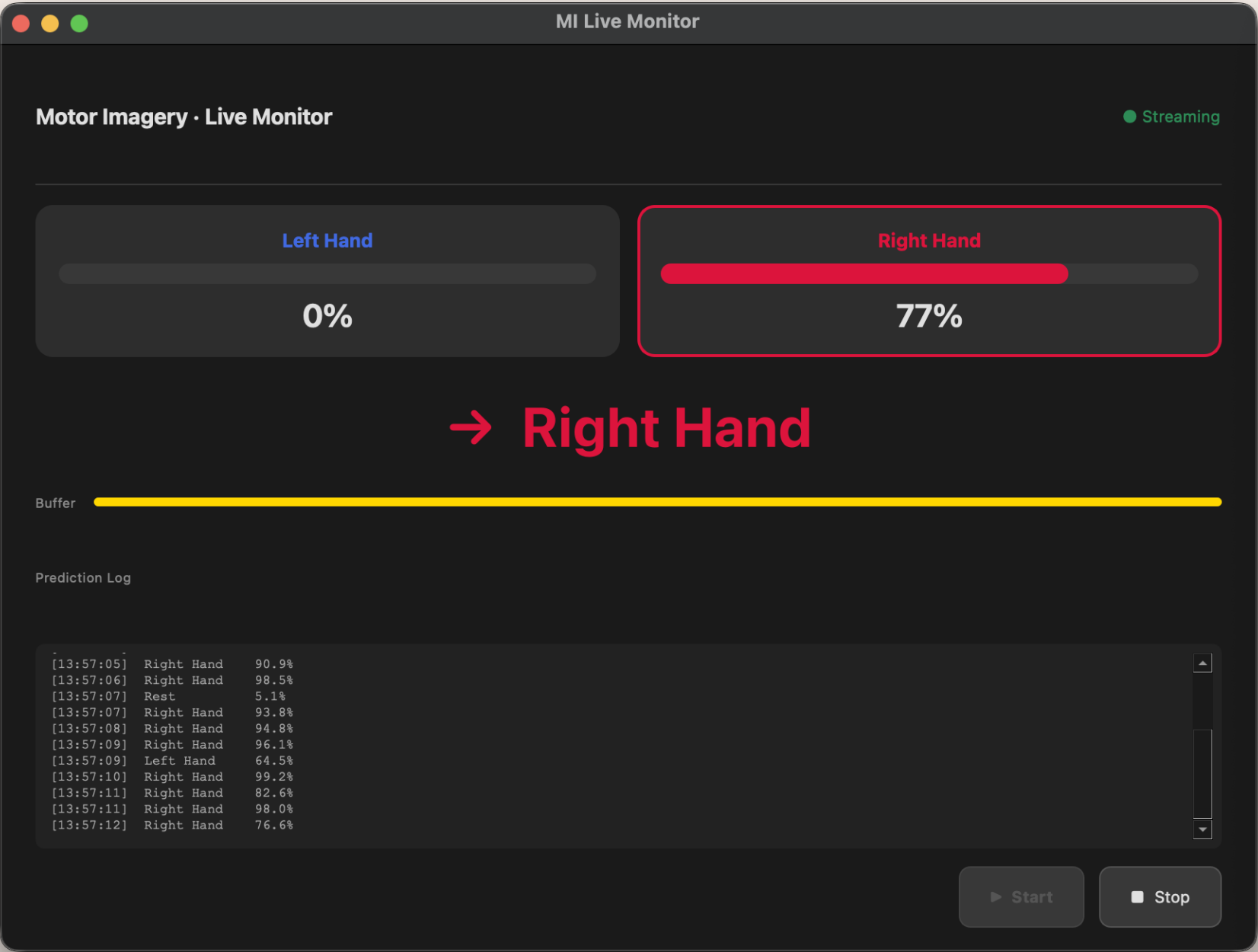
I. Finished tasks

2. Traditional Calibrating Method Design [STP11] - 100%



I. Finished tasks

2. Traditional Calibrating Method Design [STP11] - 100%



III. Current Difficulties

- The full-text deadline for BME conference
- Game idea

III. Next Plan

- Implement the game



THANK YOU
FOR READING!

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